

The properties of cryptocurrencies

Listed below are the broad features of typical cryptocurrencies.

- **Encryption:** In cryptocurrencies, two keys are created. These are linked to each other mathematically, i.e., one key is for encryption and the other for decryption. The key used for encryption becomes the public key and can be shared with the public to accept payments. The key used for decryption is the private key and must be kept secret to access the cryptocurrencies.
- **Decentralised organisation:** Cryptocurrencies function as decentralised systems. They are based on a set of protocols not governed by a single person and function on predefined mathematical laws that the entire network comes to a consensus on.
- **Distributed systems:** Also known as Distributed Ledger Technology (DLT), this is the system by which the blockchain operates across all the computers and every transaction gets verified and recorded on all of them. Everyone has access to others' ledgers and transactions get verified by the members on the network.
- **Open source:** Cryptocurrencies are open source. They can be modified and used by developers to create APIs without paying any additional fees. Developers can design apps and software for exchange and management.
- **Cryptocurrency mining:** This is a process that results in two activities – the verification of transactions and the generation of new cryptocurrencies. New currency is generated as a reward to the miners because of their efforts towards the verification process. Mining is a process that can be carried out by anyone with the required hardware devices and an Internet connection.

Cryptocurrency wallets

A cryptocurrency wallet (or crypto-wallet) is a secure digital currency wallet used to store, send and receive digital currency like Bitcoin. It is a software program to manage digital money by storing private and public keys, and interact with various blockchains to enable users to send and receive digital currency and monitor the balance. In cryptocurrency wallets, the public key performs the task of allowing other wallets to make payments to the wallet's address, whereas the private key enables the spending of cryptocurrency from the address.

How cryptocurrency wallets work

Every cryptocurrency wallet depicts a bank account number. Currently, the bank account number has to be shared with other users, to enable them to transfer funds to one's account, from other users accounts. It can be shared with an employer to transfer one's salary, with a

customer to transfer payments due to us or with any other person to whom we want to transfer money to. When anyone sends cryptocurrency to another person, they are transferring their coins' ownership to the receiver's wallet address. To be able to spend the coins, the receiver must have the private key that matches the public address the currency is assigned to. When the private and public keys match, the digital wallet balance gets increased.

In the world of cryptocurrency, if anyone wants to transfer coins to you, all you do is give them your wallet address. Just like in the real world, no two wallet addresses are ever the same, which means that there is no chance that somebody else will get your funds. Also, there is no limit to the amount of wallet addresses you can create.

Here is what a Bitcoin address looks like:

`A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa`

As you can see, it uses a combination of numbers and letters, both upper case and lower case. As most blockchains are transparent, it is easy to find out how much money a certain cryptocurrency wallet has, as well as the transactions the owner has previously made. However, a cryptocurrency wallet address does not reveal the real-world identity of its owner, which is why the blockchain is referred to as 'pseudonymous'.

Here is an example of a public and a private key.

Private key:

`03bf350d2821375158a608b51
e3e898e507fe47f2d2e8c774de4a9a7edecf74eda`

Public key:

`99b1ebcfc11a13df5161aba8160460fe1601d541`

Types of cryptocurrency wallets

There are three types of cryptocurrency wallets – software, hardware and paper based. Software wallets can be sub-categorised into three types – desktop, mobile or Web/online wallets. The different types of wallets include full node, custodial, coin-specific, network-specific, and universal or multi-coin wallets.

- **Hardware wallets:** These are termed as physical devices to store private keys offline and are connected to computers for users to log in and access the blockchain. In terms of security, hardware wallets are considered very strong. These wallets are never connected to the Internet except to transfer funds. Examples include Ledger Nano S, Trezor, Keep Key, etc.
- **Software wallets:** Termed a Bitcoin application, these